# Homework Problems

Aniruddha Madhava

Fall 2026

---

## Contents

**1 Overview** **3**

**I Functional and Harmonic Analysis** **4**

**2 Stein & Shakarchi** **4**
2.1 $L^p$ Spaces . . . 4
Problem 1.1 . . . 4
Problem 1.8 . . . 5
Problem 1.12 . . . 6
2.2 Introductory Harmonic Analysis . . . 7
Problem 2.2 . . . 7
Problem 2.3 . . . 8
Problem 2.10 . . . 9
Problem 2.16 . . . 10
Problem 2.20 . . . 11
Problem 3.1 . . . 12
Problem 3.11 . . . 12

**3 Evans** **14**
3.1 Linear Wave Equation . . . 14
Problem 2.18 . . . 14
Problem 2.19 . . . 14
Problem 2.21 . . . 15

**4 Luk** **18**
4.1 Notes . . . 18
Problem N.1 . . . 18
Problem N.2 . . . 18
Problem N.3 . . . 19
4.2 Example Sheets . . . 19
Problem E.1.5 . . . 19
Problem E.1.6 . . . 20
Problem E.1.9 . . . 22
Problem E.1.12 . . . 22

**5 Random Computations** **23**
5.1 09/23/2026 . . . . . 23
Problem 1 . . . . . 23
5.2 10/07/2026 . . . . . 25
Problem 2 . . . . . 25
Problem 3 . . . . . 25

[0]Last updated: 2026-10-10 at 17:31

# 1 Overview

The following document is a compilation of various calculations, computations, and exercises I completed as part of orals prep. The work here is not meant to be an extensive account of my work, by any means.

# Part I

# Functional and Harmonic Analysis

## 2 Stein & Shakarchi

### 2.1 $L^p$ Spaces

**Problem 1.1.** Consider $L^p = L^p(\mathbb{R}^d)$ with Lebesgue measure. Let $f_0(x) = |x|^{-\alpha}$ if $|x| < 1$, $f_0(x) = 0$ for $|x| \geq 1$; also let $f_\infty(x) = |x|^{-\alpha}$ if $|x| \geq 1$, $f_\infty(x) = 0$ when $|x| < 1$. Show that:

(a) $f_0 \in L^p$ if and only if $p\alpha < d$.

(b) $f_\infty \in L^p$ if and only if $d < p\alpha$.

(c) What happens if in the definitions of $f_0$ and $f_\infty$ we replace $|x|^{-\alpha}$ by $|x|^{-\alpha}/(\log(2/|x|))$ for $|x| < 1$, and $|x|^{-\alpha}$ by $|x|^{-\alpha}/(\log(2|x|))$ for $|x| \geq 1$?

(a) Let $p \geq 1$, and define $f_0(x) : \mathbb{R}^d \to \mathbb{R}$ to be the measurable function

$$f_0(x) = \begin{cases} |x|^{-\alpha}, & |x| < 1 \\ 0, & |x| \geq 1. \end{cases}$$

Then $f_0(x) \in L^p$ if and only if $\int_{|x|\leq 1} |f_0(x)|^p \, dx = \int_{|x|\leq 1} |x|^{-p\alpha} \, dx < \infty$. To evaluate this integral, we will use integration in polar coordinates. Using Theorem 2.49 from Folland, one has

$$\int_{|x|\leq 1} \frac{1}{|x|^{p\alpha}} \, dx = \Gamma(S^{d-1}) \cdot \int_0^1 r^{-p\alpha+d-1} \, dr.$$

By calculus in one dimension, we know that the integral on the right side is finite if and only if $-p\alpha + d - 1 > -1$, which implies $p\alpha < d$. Thus, $f_0 \in L^p$ if and only if $p\alpha < d$.

(b) Define $f_\infty(x) : \mathbb{R}^d \to \mathbb{R}$ to be measurable function

$$f_\infty(x) = \begin{cases} 0, & |x| < 1 \\ |x|^{-\alpha}, & |x| \geq 1. \end{cases}$$

Then $f_\infty(x) \in L^p$ if and only if $\int_{|x|\geq 1} |f_\infty(x)|^p \, dx = \int_{|x|\geq 1} |x|^{-p\alpha} \, dx < \infty$. Using Theorem 2.49 from Folland, one has

$$\int_{|x|\geq 1} \frac{1}{|x|^{p\alpha}} \, dx = \Gamma(S^{d-1}) \cdot \int_1^\infty r^{-p\alpha+d-1} \, dr.$$

By calculus in one dimension, we know that the integral on the right side is finite if and only if $-p\alpha + d - 1 < -1$, which implies $p\alpha > d$. Thus, $f_\infty \in L^p$ if and only if $p\alpha > d$.

(c) Now define $f_0(x) : \mathbb{R}^d \to \mathbb{R}$ to be the modified function

$$f_0(x) = \begin{cases} \dfrac{|x|^{-\alpha}}{\log(2/|x|)}, & |x| < 1 \\ 0, & |x| \geq 1. \end{cases}$$

As in (a), $f_0(x) \in L^p$ if and only if

$$\int_0^1 \frac{r^{-p\alpha+d-1}}{\log(2/r)^p} \, dr < \infty.$$

Let $k \coloneqq -p\alpha + d - 1$, and define $u = \log(2/r)$. Then the above integral transforms into

$$\int_0^1 \frac{r^{-p\alpha+d-1}}{\log(2/r)^p} \, dr = -\int_{\log(2)}^\infty \frac{2^k e^{-ku}}{u^p} \, (-2e^{-u} \, du) = 2^{k+1} \int_{\log(2)}^\infty \frac{e^{-(k+1)u}}{u^p} \, du.$$

In the limit $u \to \infty$, if $p > 1$, the integrand decays to zero provided $k + 1 \geq 0$. Thus, $f_0(x) \in L^p$ if and only if $k \geq -1$, which is equivalent to requiring $p\alpha \leq d$. If $p = 1$, then $f_0(x) \in L^p$ if and only if $p\alpha < d$. Now define the modified function $f_\infty(x) : \mathbb{R}^d \to \mathbb{R}$ by

$$f_\infty(x) = \begin{cases} 0, & |x| < 1 \\ \dfrac{|x|^{-\alpha}}{\log(2|x|)}, & |x| \geq 1. \end{cases}$$

As in (b), $f_\infty(x) \in L^p$ if and only if the integral

$$\int_1^\infty \frac{r^{-p\alpha+d-1}}{\log(2r)}\, dr < \infty.$$

Taking $u = \log(2r)$ this time, the above integral transforms into

$$\int_1^\infty \frac{r^{-p\alpha+d-1}}{\log(2r)^p}\, dr = \int_{\log(2)}^\infty \frac{2^{-k}e^{ku}}{u^p}(2^{-1}e^u\, du) = 2^{-(k+1)} \int_{\log(2)}^\infty \frac{e^{(k+1)u}}{u^p}\, du.$$

Here, we must consider two cases. First, suppose $p > 1$. Then the integrand decays to zero provided $k + 1 \leq 0$ (corresponding to $p\alpha \geq d$). Thus $f_\infty(x) \in L^p$ for $p > 1$ if and only if $p\alpha \geq d$. On the other hand, if $p = 1$, then the integral is finite if and only if $k + 1 \not\leq 0$, which corresponds to $p\alpha < d$. Thus,

$$f_\infty(x) \in L^p(\mathbb{R}^d) \text{ if and only if } \begin{cases} p\alpha \geq d, & p > 1 \\ \alpha > d, & p = 1. \end{cases}$$

**Problem 1.8.** Suppose $1 \leq p < \infty$, and that $\mathbb{R}^d$ is equipped with Lebesgue measure. Show that if $f \in L^p(\mathbb{R}^d)$, then

$$\|f(x + h) - f(x)\|_{L^p} \to 0 \quad \text{as } |h| \to 0.$$

Prove that this fails when $p = \infty$.
[Hint: By the previous exercise, the continuous functions with compact support are dense in $L^p(\mathbb{R}^d)$ for $1 \leq p < \infty$.]

Fix $1 \leq p < \infty$, and assume that $\mathbb{R}^d$ is equipped with Lebesgue measure. Further fix some $f \in L^p(\mathbb{R}^d)$. By the density of $C_c(\mathbb{R}^d)$ in $L^p(\mathbb{R}^d)$, there exists a sequence $\{g_j\}_{j=1}^\infty$ of compactly supported continuous functions so that $g_j \to f$ under the $L^p$-norm. First, we will show that for every $j$, $\|g_j(h + \cdot) - g_j\|_p \to 0$ as $h \to 0$. Without loss of generality, let $|h| \leq R$ for some $R > 0$. Then by continuity of $g_j$, $|g_j(x + h) - g_j(x)|^p \to 0$ pointwise a.e. as $|h| \to 0$. On the other hand, by continuity of $g_j$ and the extreme value theorem, one has

$$|g_j(x + h) - g_j(x)|^p \leq 2^{p-1}(|g_j(x + h)|^p + |g_j(x)|^p) \leq M\chi_{K+B_R(0)} \in L^1(\mathbb{R}^d),$$

where $M =: 2^p \cdot (\max_K |g_j(x)|)^p$. Thus, by the Dominated Convergence Theorem,

$$\int_{\mathbb{R}^d} |g_j(x + h) - g_j(x)|^p\, dx \xrightarrow{|h|\to 0} 0.$$

Now fix $\varepsilon > 0$, $j$ large enough so that $\|f - g_j\|_p^p < \varepsilon/3$, and let $\delta > 0$ so that for all $|h| < \delta$, one has $\|g_j(h + \cdot) - g_j\|_p^p < \varepsilon/3$. Then we observe that

$$\begin{aligned} \|f(h + \cdot) - f\|_p^p &= \int_{\mathbb{R}^d} |f(x + h) - f(x)|^p\, dx \\ &\leq C \cdot \left[ \int_{\mathbb{R}^d} |f(x + h) - g_j(x + h)|^p\, dx + \int_{\mathbb{R}^d} |g_j(x + h) - g_j(x)|^p\, dx + \int_{\mathbb{R}^d} |g_j(x) - f(x)|^p\, dx \right] \\ &= 2C\, \|f - g_j\|_p^p + C \cdot \|g_j(h + \cdot) - g_j\|_p^p \\ &< C\varepsilon. \end{aligned}$$

where in going from the second to the third line, we used translation invariance of the $L^p$ norms. Since the left side is independent of $\varepsilon$, taking the limit $\varepsilon \to 0$ shows that $\|f(h+\cdot) - f\|_p^p \to 0$ as $|h| \to 0$.

Now consider the case of $p = \infty$. Let $f \in L^\infty(\mathbb{R}^d)$ be the following function:

$$f(x) = \begin{cases} 1, & |x| \geq 1 \\ -1, & |x| < 1. \end{cases}$$

Then for any $|h| > 0$, one has that $\|f(h+\cdot) - f\|_{L^\infty} = 2$ so that this norm cannot converge to zero.

**Problem 1.12.** Suppose the measure space $(X, \mu)$ is separable as defined in Exercise 10. Let $1 \leq p < \infty$ and $1/p + 1/q = 1$. A sequence $\{f_n\}$ with $f_n \in L^p$ is said to converge to $f \in L^p$ **weakly** if

$$\int f_n g \, d\mu \to \int fg \, d\mu \qquad \text{for every } g \in L^q.$$

(a) Verify that if $\|f - f_n\|_{L^p} \to 0$, then $f_n$ converges to $f$ weakly.
(b) Suppose $\sup_n \|f_n\|_{L^p} < \infty$. Then, to verify weak convergence, it suffices to check the above convergence for a dense subset of functions $g$ in $L^q$.
(c) Suppose $1 < p < \infty$. Show that if $\sup_n \|f_n\|_{L^p} < \infty$, then there exists $f \in L^p$ and a subsequence $\{n_k\}$ so that $f_{n_k}$ converges weakly to $f$.

(a) Assume that $\|f - f_n\|_{L^p} \to 0$. Fix an arbitrary $g \in L^q$. Then by Hölder's inequality,

$$\left| \int f_n g \, d\mu - \int fg \, d\mu \right| \leq \int |f_n - f||g| \, d\mu \leq \|f_n - f\|_p \|g\|_q \xrightarrow{n\to 0} 0.$$

Since $g \in L^q$ was arbitrary, we conclude that $f_n$ converges to $f$ weakly.

(b) Suppose $\sup_n \|f_n\|_{L^p} < \infty$, and further assume that the above result holds for some dense subset $\mathcal{A}$ in $L^q$. Fix an arbitrary $g \in L^q$, and let $\{g_j\}_{j=1}^\infty \subset \mathcal{A}$ so that $\|g - g_j\|_q \to 0$ as $j \to \infty$. Then

$$\begin{aligned} \left| \int f_n g - \int fg \right| &= \left| \int f_n g - \int f_n g_j + \int f_n g_j - \int f g_j + \int f g_j - \int fg \right| \\ &\leq \int |f_n||g - g_j| + \left| \int f_n g_j - \int f g_j \right| + \int |f||g - g_j| \\ &\leq \|f_n\|_{L^p} \|g - g_j\|_{L^q} + \left| \int f_n g_j - \int f g_j \right| + \|f\|_{L^p} \|g - g_j\|_{L^q} \\ &\leq \sup_n \|f_n\|_{L^p} \|g - g_j\|_{L^p} + \left| \int f_n g_j - \int f g_j \right| + \|f\|_{L^p} \|g - g_j\|_{L^q}. \end{aligned}$$

Thus, taking the limit $n \to \infty$ on both sides, and using the fact that for each $j$, the second term vanishes under the limit by hypothesis, one has

$$\lim_{n\to\infty} \left| \int f_n g - \int fg \right| \leq \sup_n \|f_n\|_{L^p} \|g - g_j\|_{L^p} + \|f\|_{L^p} \|g - g_j\|_{L^q}.$$

Taking the limit $j \to \infty$ shows that $\int f_n g \to \int fg$ so that $f_n$ converges to $f$ weakly.

(c) Suppose $1 < p < \infty$, and assume $\sup_n \|f_n\|_{L^p} < \infty$. First, since $X$ was assumed to be separable, then by Exercise 10(b), one has that $L^q(X, \mu)$ is separable. So let $\{g_j\}_{j=1}^\infty \subset L^q(X, \mu)$ be a countable dense collection. We consider the sequence $\{\int f_n g_1 \, d\mu\}_{n=1}^\infty \subset \mathbb{R}$. First, it follows that this sequence is uniformly bounded: for every $n$,

$$\left| \int f_n g \, d\mu \right| \leq \int |f_n \|g|| \, d\mu \leq \|f_n\|_{L^p} \|g\|_{L^q} \leq \sup_n \|f_n\|_{L^p} \|g\|_{L^q} < \infty.$$

Thus there exists a subsequence $\{f_{n_k}^{(1)}\}_{k=1}^\infty$ so that $\{\int f_{n_k}^{(1)} g_1 \, d\mu\}_{k=1}^\infty$ converges. Now consider the sequence $\{\int f_{n_k}^{(1)} g_2 \, d\mu\}_{k=1}^\infty$. As before, this sequence is uniformly bounded so that there exists a

further subsequence $\{f_{n_k}^{(2)}\}_{k=1}^{\infty}$ such that $\{\int f_{n_k}^{(2)} g_2\}_{k=1}^{\infty}$ converges. By construction, $\{\int f_{n_k}^{(2)} g_1\}$ also converges. In this manner, for each $j$, we can obtain a subsequence $\{f_{n_k}^{(j)}\}$ so that $\{\int f_{n_k}^{(j)} g_l\}_{k=1}^{\infty}$ converges for all $\ell \le j$. Now construct the sequence $\{f_{n_k}\}_{k=1}^{\infty} = \{f_{n_j}^{(j)}\}_{j=1}^{\infty}$. By construction, $\{\int f_{n_k} g_j\}_{k=1}^{\infty}$ converges for all $j$. Now we claim that $\{\int f_{n_k} g\}_{k=1}^{\infty}$ converges for all $g \in L^q$. Let $\varepsilon > 0$ be fixed, $M \coloneqq \sup_n \|f_n\|_{L^p} < \infty$, $j$ large enough so that $\|g - g_j\|_{L^q} < \varepsilon/(3M)$, and $N$ large enough so that for all $k, m \ge N$, $|\int f_{n_k} g_j - \int f_{n_m} g_j| < \varepsilon/3$. Then for all $k, m \ge N$, one has

$$\begin{aligned}\left|\int f_{n_k} g - \int f_{n_m} g\right| &\le 2 \sup_n \|f_n\|_{L^p} \cdot \|g - g_j\|_{L^q} + \left|\int f_{n_k} g_j - \int f_{n_m} g_j\right| \\ &< \frac{2\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon.\end{aligned}$$

Thus taking $\varepsilon \to 0$ concludes the proof of the above claim. Now define the linear functional $T : L^q \to \mathbb{R}$ as follows: for any $g \in L^q$, let

$$T(g) \coloneqq \lim_{n_j \to \infty} \int f_{n_j} g.$$

It is straightforward to see that this funtional is bounded. Thus since $1 < p, q < \infty$ and $p, q$ are conjugates, by the Riesz Representation Theorem, there exists a unique $f \in L^p$ so that

$$\int fg = T(g) = \lim_{n_j \to \infty} \int f_{n_j} g \qquad (g \in L^q(X, \mu)).$$

Thus the proof concludes.

## 2.2 Introductory Harmonic Analysis

**Problem 2.2.** The following are simple generalizations of the Hausdorff-Young inequalities.

(a) Suppose $\{\varphi_n\}$ is an orthonormal sequence on $L^2(X, \mu)$. Assume also that $|\varphi_n(x)| \le M$ for all $n$. If $a_n = \int f \bar{\varphi}_n \, d\mu$, then $\|a_n\|_{L^q} \le M^{(2/p)-1} \|f\|_{L^p(X)}$, $1 \le p \le 2$, $1/p + 1/q = 1$.

(b) Suppose $f \in L^p$ on the torus $\mathbb{T}^d$, and $a_n = \int_{\mathbb{T}^d} f(x) e^{-2\pi i n \cdot x} \, dx$, $n \in \mathbb{Z}^d$. Then $\|\{a_n\}\|_{L^q(\mathbb{Z}^d)} \le \|f\|_{L^p(\mathbb{T}^d)}$, where $1/q \le 1 - 1/p$.

(a) We initially define the operator $T$ on simple functions $f$ by $Tf = \{a_n\}$, where

$$a_n \coloneqq \int f \bar{\varphi}_n \, d\mu.$$

Then clearly, for $(p_0, q_0) = (1, \infty)$, one has that since for each $n$,

$$|a_n| \le \int |f| |\bar{\varphi}_n| \, d\mu \le \int M |f| = M \|f\|_{L^1(X)},$$

$\|a_n\|_{L^\infty} = \sup |a_n| \le M \|f\|_{L^1(X)}$. Thus by Proposition 5.4, we conclude that there exists a unique extension $T$ to all of $L^1(X)$ satisfying the same bound. Likewise, for $(p_1, q_1) = (2, 2)$ and any simple function $f$, Bessel's inequality implies that

$$\|a_n\|_{L^2} = \left(\sum_{n=1}^{\infty} |a_n|^2\right)^{1/2} \le \|f\|_{L^2(X)} = M^{\frac{2}{2}-1} \|f\|_{L^2(X)}.$$

Thus by Proposition 5.4, there exists a unique extension $T$ to all of $L^2(X)$ satisfying the same bound. Then, the Riesz-Thorin Interpolation Theorem tells us that $\|a_n\|_{L^q} \le M \|f\|_{L^p(X)}$ where

$$\frac{1}{p} = \frac{1-t}{p_0} + \frac{t}{p_1} = 1 - \frac{1}{2}t, \qquad \frac{1}{q} = \frac{1-t}{q_0} + \frac{t}{q_1} = \frac{1}{2}t,$$

where $t \in [0, 1]$ (thus corresponding to $1 \le p \le 2$, $p^{-1} + q^{-1} = 1$), and where

$$M \le M^{1-t} = M^{2(1-(1/2)t)-1} = M^{\frac{2}{p}-1}.$$

(b) As before, initially define the operator $T$ on simple functions $f$ on $\mathbb{T}^d$ by $Tf = \{a_n\}$, where for each $n \in \mathbb{Z}^d$,

$$a_n \coloneqq \int_{\mathbb{T}^d} f(x) e^{-2\pi i n \cdot x}\, dx.$$

Clearly for $(p_0, q_0) = (1, \infty)$, one has

$$|a_n| \le \int_{\mathbb{T}^d} |f(x)| = \|f\|_{L^1(\mathbb{T}^d)},$$

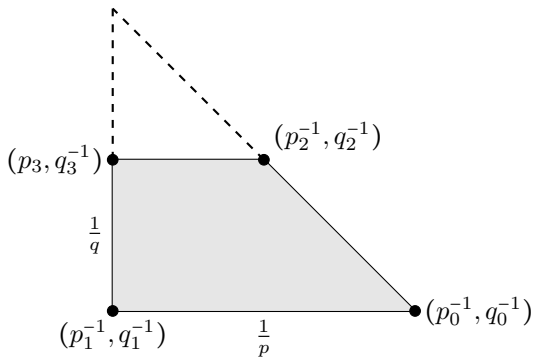
so that

$$\|\{a_n\}\|_{L^\infty(\mathbb{Z}^d)} = \sup_n |a_n| \le \|f\|_{L^1(\mathbb{T}^d)}.$$

Thus Proposition 5.4 gives a unique extension of $T$ to all $L^1(\mathbb{T}^d)$ that satisfies the same bound. Now consider $(p_1, q_1) = (\infty, \infty)$. Then for any $n \in \mathbb{Z}^d$, one has that

$$|a_n| \le \int_{\mathbb{T}^d} |f(x)| \le \|f\|_{L^\infty(\mathbb{T}^d)},$$

where we assume that $dx$ denotes the normalized measure on $\mathbb{T}^d$. Thus taking the supremum over all $n \in \mathbb{Z}^d$, and using Proposition 5.4 concludes the claim. Likewise, one can show this property holds for the pair $(p_2, q_2) = (2, 2)$. Since $\mathbb{T}^d$ has finite measure, it follows that for all $p \ge 2$ (i.e., $1/p \le 1/2$), $L^p(\mathbb{T}^d) \hookrightarrow L^2(\mathbb{T}^d)$. Thus, using Proposition 5.4, we also get the endpoint $(p_3, q_3) = (p, 2)$ for all $p \ge 2$. The Riesz diagram as of now is given by




**Problem 2.3.** Check that an inequality of the form $\|\hat{f}\|_{L^q(\mathbb{R}^d)} \le A\|f\|_{L^p(\mathbb{R}^d)}$ (holding for all simple functions $f$) is possible if and only if $1/p + 1/q = 1$.
[Hint: Let $f_r(x) = f(rx)$, $r > 0$. Then $\hat{f}_r(\xi) = \hat{f}(\xi/r) r^{-d}$.]

Let $f$ be an arbitrary fixed simple function. As suggested by the hint, let $r > 0$, and define $f_r(x) = f(rx)$ so that $\hat{f}_r(\xi) = \hat{f}(\xi/r) r^{-d}$. We note that $f_r$ is also a simple function. We now compute,

$$\begin{aligned}
\|\hat{f}_r\|_{L^q(\mathbb{R}^d)} &= \left( \int_{\mathbb{R}^d} |\hat{f}(\xi/r)|^q r^{-dq}\, d\xi \right)^{1/q} \\
&= r^{-d} \left( \int_{\mathbb{R}^d} |\hat{f}(\xi/r)|^q\, d\xi \right)^{1/q} \\
&= r^{-d} \left( \int_{\mathbb{R}^d} |\hat{f}(\eta)|^q r^d\, d\eta \right)^{1/q} \\
&= r^{d(-1+1/q)} \|\hat{f}\|_{L^q(\mathbb{R}^d)}.
\end{aligned}$$

Likewise,

$$\begin{aligned}\|f_r\|_{L^p(\mathbb{R}^d)} &= \left(\int_{\mathbb{R}^d} |f(rx)|^p \, dx\right)^{1/p} \\ &= \left(\int_{\mathbb{R}^d} |f(y)| r^{-d} \, dy\right)^{1/p} = r^{-d/p} \|f\|_{L^p(\mathbb{R}^d)}.\end{aligned}$$

So we have the following: the above estimate holds for all simple functions $f$ if and only if for any $r > 0$,

$$\left\|\hat{f}_r\right\|_{L^q(\mathbb{R}^d)} \le A \|f\|_{L^p(\mathbb{R}^d)} \quad \text{and} \quad \left\|\hat{f}\right\|_{L^q(\mathbb{R}^d)} \le A \|f\|_{L^p(\mathbb{R}^d)}.$$

But our work above shows that the first inequality is equivalent to the estimate

$$r^{-d+d/q} \left\|\hat{f}\right\|_{L^q(\mathbb{R}^d)} \le r^{-d/p} \|f\|_{L^p(\mathbb{R}^d)} \iff \left\|\hat{f}\right\|_{L^q(\mathbb{R}^d)} \le r^{d(1-1/q-1/p)} A \|f\|_{L^q(\mathbb{R}^d)}.$$

Thus since $r > 0$ is arbitrary, the above estimate is consistent with the two inequality we have above if and only if $1 - 1/q - 1/p = 0$ iff $1/q + 1/p = 1$.

**Problem 2.10.** Suppose $f \in L^p(\mathbb{R})$. Verify that:
(a) $\|f * \mathcal{P}_y\|_{L^p(\mathbb{R})} \le \|f\|_{L^p}$, $1 \le p \le \infty$.
(b) $f * \mathcal{P}_y \to f$, as $y \to 0$ in the $L^p$ norm when $1 \le L^p < \infty$.

(a) Let $\mathcal{P}_y(x)$ denote the **Poisson kernel**,

$$\mathcal{P}_y(x) \coloneqq \frac{1}{\pi} \frac{y}{x^2 + y^2}.$$

Let $p_0 = 1$, and define the operator $T$ on $L^{p_0}(\mathbb{R})$, initially just on the set of all simple functions, by $Tf \coloneqq (f * \mathcal{P}_y)(x)$. First, if $y = 0$, then $P_y(x) \equiv 0$ so that $Tf \equiv 0$, and the result holds trivially. So assume $y \neq 0$ (wlog, assume $y > 0$). Then we compute,

$$\begin{aligned}\|\mathcal{P}_y\|_{L^1(\mathbb{R})} &= \int_{\mathbb{R}} \frac{1}{\pi} \frac{|y|}{x^2 + y^2} \, dx \\ &= \frac{1}{\pi} \frac{|y|}{y} \arctan\left(\frac{x}{y}\right)\Big|_{-\infty}^{\infty} = \frac{1}{\pi} \frac{|y|}{y} \cdot \left[\frac{\pi}{2} + \frac{\pi}{2}\right] = \frac{|y|}{y} = 1.\end{aligned}$$

Thus for $p_0 = 1$, we have

$$\begin{aligned}\|Tf\|_{L^1(\mathbb{R})} &= \int_{\mathbb{R}} \left| \frac{1}{\pi} \int_{\mathbb{R}} \frac{f(x-t)y}{y^2 + t^2} \, dt \right| dx \\ &\le \int_{\mathbb{R}} \int_{\mathbb{R}} |f(x-t)| \cdot \left| \frac{1}{\pi} \frac{y}{y^2 + t^2} \right| dx dt \\ &= \int_{\mathbb{R}} \|f(\cdot - t)\|_{L^1(\mathbb{R})} \cdot \left| \frac{1}{\pi} \frac{y}{y^2 + t^2} \right| dt \\ &= \int_{\mathbb{R}} \|f\|_{L^1(\mathbb{R})} \cdot \left| \frac{1}{\pi} \frac{y}{y^2 + t^2} \right| dt = \|f\|_{L^1(\mathbb{R})} \|\mathcal{P}_y\|_{L^1(\mathbb{R})} = \|f\|_{L^1(\mathbb{R})},\end{aligned}$$

where the second line follows from Minkowski's inequality, and the fourth line follows from translation invariance of the $L^p$ norms. Thus by Proposition 5.4, $T$ extends to a unique operator on all of $L^1(\mathbb{R})$ satisfying the same estimate. Now define the operator $T$ on $L^{p_1}(\mathbb{R})$, where $p_1 = \infty$, initially on just the set of all simple functions by $Tf \coloneqq (f * \mathcal{P}_y)(x)$. As before, the proof is trivial if $y = 0$. So assume

$y \neq 0$. Then for each $x$,

$$\begin{aligned}
|Tf(x)| &= \left| \int_{\mathbb{R}} \frac{1}{\pi} \frac{f(x-t)y}{y^2+t^2}\, dt \right| \\
&\leq \int_{\mathbb{R}} |f(x-t)| \cdot \left| \frac{1}{\pi} \frac{y}{y^2+t^2} \right| dt \\
&\leq \int_{\mathbb{R}} \|f(x-\cdot)\|_{L^\infty(\mathbb{R})} \cdot \left| \frac{1}{\pi} \frac{y}{t^2+y^2} \right| dt \\
&= \int_{\mathbb{R}} \|f\|_{L^\infty(\mathbb{R})} \cdot \left| \frac{1}{\pi} \frac{y}{t^2+y^2} \right| dt = \|f\|_{L^\infty(\mathbb{R})} \cdot \|\mathcal{P}_y\|_{L^1(\mathbb{R})} = \|f\|_{L^\infty(\mathbb{R})} .
\end{aligned}$$

Using Proposition 5.4, the result extends to all of $L^\infty(\mathbb{R})$. On the Riesz diagram, we have proven the estimate for the endpoints $(0,0)$ and $(1,1)$. Thus by the Riesz-Thorin interpolation theorem, the theorem follows for all points on the diagonal line connecting these endpoints (i.e., for all $1 \leq p \leq \infty$).

(b) Now let $1 \leq p < \infty$, and let $\delta > 0$ be fixed. By continuity of translations in $L^p(\mathbb{R})$, there exists $\varepsilon > 0$ such that $|t| \leq \delta \implies \|f(\cdot - t) - f\|_{L^p(\mathbb{R})} < \varepsilon/2$. Thus we compute,

$$\begin{aligned}
\|f * \mathcal{P}_y - f\|_{L^p} &= \left( \int_{\mathbb{R}} \left| \int_{\mathbb{R}} \frac{1}{\pi} \frac{f(x-t)y}{t^2+y^2}\, dt - f(x) \right|^p dx \right)^{1/p} \\
&= \left( \int_{\mathbb{R}} \left| \int_{\mathbb{R}} [f(x-t) - f(x)] \cdot \frac{1}{\pi} \frac{y}{t^2+y^2}\, dt \right|^p dx \right)^{1/p} \\
&\leq \int_{\mathbb{R}} \left( \int_{\mathbb{R}} |f(x-t) - f(x)| \cdot \left| \frac{1}{\pi} \frac{y}{t^2+y^2} \right|^p dx \right)^{1/p} dt \\
&= \int_{\mathbb{R}} \|f(\cdot - t) - f\|_{L^p(\mathbb{R})} \cdot \left| \frac{1}{\pi} \frac{y}{t^2+y^2} \right| dt \\
&= \int_{|t|\leq\delta} \|f(\cdot - t) - f\|_{L^p(\mathbb{R})} \cdot \left| \frac{1}{\pi} \frac{y}{t^2+y^2} \right| dt + \int_{|t|>\delta} \|f(\cdot - t) - f\|_{L^p(\mathbb{R})} \cdot \left| \frac{1}{\pi} \frac{y}{t^2+y^2} \right| dt \\
&\leq \varepsilon/2 + 2\|f\|_{L^p(\mathbb{R})} \cdot \int_{|t|>\delta} |\mathcal{P}_y(t)|\, dt,
\end{aligned}$$

where the second line follows from $\|\mathcal{P}_y\|_{L^1(\mathbb{R})} = 1$ for any $y \in \mathbb{R}$, and the third line follows from Minkowski's inequality. Since $\mathcal{P}_y(t) \to 0$ for all $t \neq 0$ in the limit $y \to 0^+$, it follows that the second term tends to 0 in the limit $y \to 0^+$. I.e., there exists some $y_0$ so that for all $y < y_0$, the second term is bounded above by $\varepsilon/(2\|f\|_{L^p(\mathbb{R})})$. Therefore, we conclude that $\|f * \mathcal{P}_{y_0} - f\|_{L^p(\mathbb{R})} \leq \varepsilon$ so that $f * \mathcal{P}_y \to f$ in the $L^p$ norm.

**Problem 2.16.** Prove that the space $\mathbf{H}^1_r$ is complete as follows. Let $\{f_n\}$ be a Cauchy sequence in $\mathbf{H}^1_r$. Then since $\{f_n\}$ is also Cauchy in $L^1$, there is an $L^1$ function $f$ so that $f = \lim_{n\to\infty} f_n$ in the $L^1$-norm. Now for an appropriate subsequence $\{n_k\}$, write $f = f_{n_1} + \sum_{k=1}^{\infty} (f_{n_{k+1}} - f_{n_k})$.

Let $\mathbf{H}^1_r$ denote the Hardy space, and let $\{f_n\} \subset \mathbf{H}^1_r$ be a Cauchy sequence. First, we begin by showing that each $f \in L^1$. Let $\sum_j \lambda_j \mathfrak{a}_j$ be an arbitrary decomposition of $f$, where each $\mathfrak{a}_j$ is an atom, and $\sum |\lambda_j| < \infty$. Thus since the sum converges under the $L^1$ norm, Fubini/Tonelli implies that

$$\int_{\mathbb{R}} |f|\, dx = \int_{\mathbb{R}} \left| \sum_1^\infty \lambda_k \mathfrak{a}_k \right| \leq \int_{\mathbb{R}} \sum_1^\infty |\lambda_k||\mathfrak{a}_k| = \sum_1^\infty \int_{\mathbb{R}} |\lambda_k||\mathfrak{a}_k| = \sum_1^\infty |\lambda_k| < \infty.$$

Thus since $\sum \lambda_j \mathfrak{a}_j$ was an arbitrary decomposition of $f$ into a sum of atoms, it follows that $f \in L^1$. In particular, taking the infimum over all such decompositions also yields the estimate $\|f\|_{L^1} \leq \|f\|_{\mathbf{H}^1_r}$. Thus if $\{f_n\}$ is Cauchy in $\mathbf{H}^1_r$, they are also Cauchy in $L^1$. Since the latter is a Banach space, there exists an $L^1$ function $f$ such that $f = \lim_{n\to\infty} f_n$ in the $L^1$-norm. But this implies there exists a subsequence $\{f_{n_k}\}$ that

converges to $f$ pointwise a.e. Fix $\varepsilon > 0$. Since $\{f_n\}$ is Cauchy in $\mathbf{H}^1_r$, for every $k \geq 1$, there exists a positive integer $N_k$ such that for all $m, n > N_k$,

$$\|f_m - f_n\|_{\mathbf{H}^1_r} < \frac{\varepsilon}{2^k}.$$

Thus from the subsequence $\{f_{n_k}\}_{k=1}^{\infty}$, we construct another subsequence as follows: we choose $n_1 > N_1$. Next, we choose $n_2 > \max\{n_1, N_2\}$. Repeating inductively, we obtain a subsequence $\{f_{n_k}\}$ such that

$$\sum_{k=1}^{\infty} \|f_{n_{k+1}} - f_{n_k}\|_{\mathbf{H}^1_r} \leq \sum_{k=1}^{\infty} \frac{\varepsilon}{2^k} = \varepsilon < \infty.$$

Now we recall that

$$f = \lim_{j\to\infty} f_{n_j} = f_{n_1} + \lim_{j\to\infty} \sum_{k=1}^{j} (f_{n_{k+1}} - f_{n_k}) = f_{n_1} + \sum_{1}^{\infty} (f_{n_{k+1}} - f_{n_k}).$$

Thus

$$\|f\|_{\mathbf{H}^1_r} \leq \|f_{n_1}\|_{\mathbf{H}^1_r} + \sum_{k=1}^{\infty} \|f_{n_{k+1}} - f_{n_k}\|_{\mathbf{H}^1_r} < \infty,$$

which estalishes that $f \in \mathbf{H}^1_r$. All that remains to be shown is that $f_n \to f$ under the $\mathbf{H}^1_r$ norm. We compute,

$$\|f - f_n\|_{\mathbf{H}^1_r} \leq \|f - f_{n_k}\|_{\mathbf{H}^1_r} + \|f_{n_k} - f_n\|_{\mathbf{H}^1_r}.$$

Since the sequence $\{f_n\}$ is Cauchy in $\mathbf{H}^1_r$, the second term vanishes in the limit $n, k \to \infty$. On the other hand, since

$$\|f - f_{n_k}\|_{\mathbf{H}^1_r} \leq \sum_{j=k}^{\infty} \|f_{n_{k+1}} - f_{n_k}\|_{\mathbf{H}^1_r}.$$

By our construction of the subsequence $\{f_{n_k}\}$, we can control the sum by $\varepsilon$. Thus we conclude that $f_n \to f$ under the $\mathbf{H}^1_r$ norm, which means that $\mathbf{H}^1_r$ is a Banach space.

**Problem 2.20.** When $f \in L^1(\mathbb{R}^d)$ we know that its Fourier transform is bounded and $\hat{f}(\xi)$ tends to 0 as $|\xi| \to \infty$ (the Riemann-Lebesgue lemma), but no better assertion about the "smallness" of $\hat{f}$ can be made. (For the analogous result for Fourier series, see Chapter 3 in Book III.) Show, however, that for $f \in \mathbf{H}^1_r$ we have

$$\int_{\mathbb{R}^d} |\hat{f}(\xi)| \, \frac{d\xi}{|\xi|^d} \leq A \|f\|_{\mathbf{H}^1_r}.$$

[Hint: Verify this for atoms.]

First, assume that for every atom $\mathfrak{a}$, the above estimate holds Then for any decomposition, $f = \sum_{k=1}^{\infty} \lambda_k \mathfrak{a}_k$ into atoms,

$$\begin{aligned}
\int_{\mathbb{R}^d} |\hat{f}(\xi)| \, \frac{d\xi}{|\xi|^d} &= \int_{\mathbb{R}^d} \left| \sum_{1}^{\infty} \lambda_k \hat{\mathfrak{a}}_k \right| \, \frac{d\xi}{|\xi|^d} \\
&\leq \sum_{1}^{\infty} \int_{\mathbb{R}^d} |\lambda_k| |\hat{\mathfrak{a}}_k| \, \frac{d\xi}{|\xi|^d} \\
&\leq A \sum_{1}^{\infty} |\lambda_k|.
\end{aligned}$$

Thus taking the infimum over all such decompositions of $f$, we obtain,

$$\int_{\mathbb{R}^d} |\hat{f}(\xi)| \, \frac{d\xi}{|\xi|^d} \lesssim \|f\|_{\mathbf{H}^1_r}.$$

So now, we will show that the result holds true for every atom. Let $\mathfrak{a}$ be an atom on $\mathbb{R}^d$; without loss of generality, we can assume that $\mathfrak{a}$ is supported on the unit ball $B_1(0) \subset \mathbb{R}^d$. Assume $\mathfrak{a}$ is not identically zero. Then one has that

$$\hat{\mathfrak{a}}(\xi) \coloneqq \int_{\mathbb{R}^d} \mathfrak{a}(x) e^{-2\pi i x\cdot\xi}\, dx = \int_{|x|\le 1} \mathfrak{a}(x) e^{-2\pi i x\cdot\xi}\, dx = \int_{|x|\le 1} \mathfrak{a}(x)\left(e^{-2\pi i x\cdot\xi} - 1\right)\, dx,$$

where the final equality follows from the fact that $\int_{\mathbb{R}^d} \mathfrak{a}(x) = 0$. Thus,

$$\begin{aligned}
\int_{\mathbb{R}^d} |\hat{\mathfrak{a}}(\xi)|\, \frac{d\xi}{|\xi|^d} &= \int_{|\xi|\le 1} \left| \int_{|x|\le 1} \mathfrak{a}(x) \frac{e^{-2\pi i x\cdot\xi} - 1}{|\xi|^d}\, dx \right| d\xi + \int_{|\xi|>1} |\hat{\mathfrak{a}}(\xi)|\, \frac{d\xi}{|\xi|^d} \\
&\le \int_{|x|\le 1} \int_{|\xi|\le 1} |\mathfrak{a}(x)| \cdot \left| \frac{e^{-2\pi i x\cdot\xi} - 1}{|\xi|^d} \right| d\xi dx + \int_{|\xi|>1|} |\hat{\mathfrak{a}}(\xi)|\, \frac{d\xi}{|\xi|^d} \\
&\le 2\pi \|\mathfrak{a}\|_{L^1(\mathbb{R}^d)} + \left( \int_{|\xi|>1} |\hat{\mathfrak{a}}(\xi)|^2\, d\xi \right)^{1/2} \left( \int_{|\xi|>1} \frac{1}{|\xi|^{2d}}\, d\xi \right)^{1/2} \\
&= 2\pi \|\mathfrak{a}\|_{L^1(\mathbb{R}^d)} + \left( \int_{|x|\le 1} |\mathfrak{a}(\xi)|^2\, dx \right)^{1/2} \\
&\le 2\pi \|\mathfrak{a}\|_{L^1(\mathbb{R}^d)} + \left( \frac{1}{m(B_1(0))} \right)^{1/2} \le 2\pi + \frac{1}{m(B_1(0))^{1/2}} \equiv A < \infty.
\end{aligned}$$

**Problem 3.1.** Suppose $F$ is a distribution on $\Omega$ and $F = f$, with $f$ a $C^k$ function in $\Omega$. Show that $\partial_x^\alpha F$, taken in the sense of distributions, agrees with $\partial_x^\alpha f$ for each $|\alpha| \le k$.

Let $\Omega \subseteq \mathbb{R}^d$, and let $F$ be a distribution on $\Omega$ such that $F = f$, where $f \in C^k(\Omega)$. Recall that we defined $\partial_x^\alpha F$ as the distribution,

$$(\partial_x^\alpha F)(\varphi) \coloneqq (-1)^{|\alpha|} F(\partial_x^\alpha \varphi), \qquad \forall \varphi \in C_c^\infty(\Omega).$$

When $|\alpha| = 1$, we compute,

$$\begin{aligned}
\int_\Omega (\partial_{x_j} f)\varphi\, dx &= f\varphi\Big|_{\partial\Omega} - \int_\Omega f \partial_{x_j}\varphi\, dx \\
&= -\int_\Omega f \partial_{x_j}\varphi\, dx = -F(\partial_{x_j}\varphi),
\end{aligned}$$

where the boundary term is zero since $\operatorname{supp}(\varphi) \subset \Omega$. Now for each $1 \le |\alpha| \le k$, applying induction on the above steps (by "swapping" one derivative at a time), we obtain,

$$\int_\Omega (\partial_x^\alpha f)\varphi\, dx = (-1)^{|\alpha|} \int_\Omega f \partial_x^\alpha \varphi\, dx =: (-1)^{|\alpha|} F(\partial_x^\alpha \varphi),$$

for all $\varphi \in C_c^\infty(\Omega)$.

**Problem 3.11.** Suppose that $\varphi_1, \varphi_2 \in \mathcal{S}$.
(a) Verify that $\varphi_1 \cdot \varphi_2$ belongs to $\mathcal{S}$.
(b) Using the Fourier transform, prove that $\varphi_1 * \varphi_2 \in \mathcal{S}$.
(c) Show directly from the definition of convolution that $\varphi_1 * \varphi_2 \in \mathcal{S}$.

Let $\varphi_1, \varphi_2 \in \mathcal{S}$.

(a) Let $N$ be an arbitrary fixed positive integer. For any $x \in \mathbb{R}^d$, and $|\alpha|, |\beta| \le N$, one has that

$$x^\beta \partial_x^\alpha (\varphi_1 \cdot \varphi_2)(x) = x^\beta \sum_{\gamma \le \alpha} \binom{\alpha}{\gamma} (\partial^\gamma \varphi_1)(\partial^{\alpha-\gamma}\varphi_2).$$

This means that

$$\begin{aligned}|x^\beta \partial_x^\alpha(\varphi_1\cdot\varphi_2)(x)| &\le \sum_{\gamma\le\alpha}\binom{\alpha}{\gamma}\left|x^\beta(\partial_x^\gamma\varphi_1)(x)(\partial_x^{\alpha-\gamma}\varphi_2)(x)\right| \\ &\le \sum_{\gamma\le\alpha}\binom{\alpha}{\gamma}|x^\beta(\partial_x^\gamma\varphi_1(x))|\,\|\partial_x^{\alpha-\gamma}\varphi_2\|_{L^\infty(\mathbb{R}^d)}.\end{aligned}$$

Thus taking the supremum over all $x\in\mathbb{R}^d$ and $|\alpha|,|\beta|\le N$ gives $\|\varphi_1\cdot\varphi_2\|_N<\infty$. Since this is true for all positive integers $N$, the proof concludes.

(b) Since $\varphi_{1,2}$ are Schwartz, both of their Fourier transforms are also Schwartz. Thus by (a), $\hat\varphi_1\cdot\hat\varphi_2$ is also Schwartz. But $\hat\varphi_1\cdot\hat\varphi_2=(\varphi_1*\varphi_2)\hat{}$. Therefore, $(\varphi_1*\varphi_2)\hat{}\in\mathcal{S}$. Since the inverse Fourier transform also takes Schwartz functions to Schwartz functions, we conclude that $\varphi_1*\varphi_2$ is Schwartz.

(c) Recall that

$$(\varphi_1*\varphi_2)(x)=\int_{\mathbb{R}^d}\varphi_1(x-y)\varphi_2(y)\,dy.$$

Therefore, for any $x\in\mathbb{R}^d$, fixed positive integer $N$, and $|\alpha|,|\beta|\le N$,

$$|x^\beta\partial_x^\alpha(\varphi_1*\varphi_2)(x)|=\left|x^\beta\int_{\mathbb{R}^d}\partial_x^\alpha\varphi_1(x-y)\varphi_2(y)\,dy\right|\le\int_{\mathbb{R}^d}|x^\beta\partial_x^\alpha\varphi_1(x-y)||\varphi_2(y)|\,dy$$

# 3 Evans

## 3.1 Linear Wave Equation

**Problem 2.18.** (Stokes' rule) Assume $u$ solves the initial-value problem

$$\begin{cases} u_{tt} - \Delta u = 0 & \text{in } \mathbb{R}^n \times (0,\infty) \\ u = 0, u_t = h, & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{cases}$$

Show that $v \coloneqq u_t$ solves

$$\begin{cases} v_{tt} - \Delta v = 0 & \text{in } \mathbb{R}^n \times (0,\infty) \\ v = h, v_t = 0, & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{cases}$$

This is *Stokes' rule.*

Assume $u$ solves the given IVP, and let $v \coloneqq u_t$. One immediately notes that on $\mathbb{R}^n \times \{t = 0\}$, since $u_t = h$, $v = h$. Second, $v_t = u_{tt}$. On $\mathbb{R}^n \times \{t = 0\}$, $u \equiv 0$ so that $\Delta u = 0$. But since $u$ satisfies the wave equation $u_{tt} = \Delta u \equiv 0$. Thus $v_t \equiv 0$ on $\mathbb{R}^n \times \{t = 0\}$. Next we note that $\partial_t$ and $\partial_{x^j}$ commute for any $1 \le j \le n$. Thus since $u$ satisfies $u_{tt} - \Delta u = 0$, differentiating both sides by $t$, one obtains

$$u_{ttt} - \Delta u_t = 0 \iff v_{tt} - \Delta v = 0$$

in $\mathbb{R}^n \times (0,\infty)$.

**Problem 2.19.**

(a) Show the general solution of the PDE $u_{xy} = 0$ is

$$u(x,y) = F(x) + G(y)$$

for arbitrary functions $F, G$.

(b) Using the change of variables $\xi = x + t$, $\eta = x - t$, show $u_{tt} - u_{xx} = 0$ if and only if $u_{\xi\eta} = 0$.

(c) Use (a) and (b) to rederive d'Alembert's formula.

(d) Under what conditions on the initial data $g, h$ is the solution $u$ a right-moving wave? A left-moving wave?

(a) Consider the PDE $u_{xy} = 0$. Let $v(x,y) = u_x$. Then $v$ satisfies the differential equation $v_y = 0$. Thus since $v$ is independent of $y$, one has $u_x(x,y) = v(x,y) = f(x)$ for some arbitrary function $f$. But then integrating both sides with respect to $x$, and supposing that $F(x)$ is the antiderivative of $f(x)$,

$$u(x,y) = F(x) + G(y),$$

where $G$ is some arbitrary function of $y$.

(b) Define $\xi = x + t$ and $\eta = x - t$ so that $x = (\xi + \eta)/2$ and $t = (\xi - \eta)/2$. Then,

$$\begin{aligned} u_\xi &= u_x \frac{\partial x}{\partial \xi} + u_t \frac{\partial t}{\partial \xi} \\ &= \frac{1}{2} u_x + \frac{1}{2} u_t. \\ u_{\xi\eta} &= \frac{1}{2}\left[u_{xx}\frac{\partial x}{\partial \eta} + u_{xt}\frac{\partial t}{\partial \eta}\right] + \frac{1}{2}\left[u_{tx}\frac{\partial x}{\partial \eta} + u_{tt}\frac{\partial t}{\partial \eta}\right] \\ &= \frac{1}{2}[u_{xx} - u_{xt}] + \frac{1}{2}[u_{tx} - u_{tt}] = \frac{1}{2}[u_{tt} - u_{xx}], \end{aligned}$$

where we used the fact that $u_{xt} = u_{tx}$. Thus, the proof concludes.

(c) Consider the initial value problem,

$$\begin{cases} u_{tt} - u_{xx} = 0 & (\text{in } \mathbb{R} \times (0,\infty)) \\ u = g, u_t = h & (\text{on } \mathbb{R} \times \{t = 0\}). \end{cases}$$

Define the change of variables $\xi = x+t$ and $\eta = x-t$ as above. By (b), $u_{tt} - u_{xx} = 0$ if and only if $u_{\xi\eta} = 0$. By (a), it follows that $u$ must be of the form,

$$u(\xi, \eta) = F(\xi) + G(\eta).$$

Thus, $u(x,t) = F(x+t) + G(x-t)$. Now we must match the initial conditions. At $t = 0$,

$$u(x,0) = F(x) + G(x) = g(x).$$
$$u_t(x,0) = F'(x) - G'(x) = h(x).$$

Differentiating the first equation, one first obtains the system

$$\begin{cases} F'(x) + G'(x) = g'(x) \\ F'(x) - G'(x) = h(x). \end{cases}$$

Thus, we get

$$F'(x) = \frac{1}{2}(g'(x) + h(x)) \implies F(x+t) = \frac{1}{2}[g(x+t) - g(0)] + \frac{1}{2}\int_0^{x+t} h(y)\, dy + F(0).$$

Likewise, we get

$$G(x-t) = \frac{1}{2}[g(x-t) - g(0)] - \int_0^{x-t} h(y)\, dy + G(0).$$

Adding the two equations, and using the fact that $F(0) + G(0) = g(0)$, we conclude that

$$u(x,t) = \frac{1}{2}[g(x-t) + g(x+t)] + \frac{1}{2}\int_{x-t}^{x+t} h(y)\, dy,$$

which is precisely d'Alembert's formula.

(d) To get a left-moving wave, we want the functions of $(x-t)$ to cancel out. Suppose $h(x) = g'(x)$. Then the fundamental theorem of calculus implies that

$$u(x,t) = \frac{1}{2}[g(x-t) + g(x+t)] + \frac{1}{2}\int_{x-t}^{x+t} g'(y)\, dy = \frac{1}{2}[g(x-t) + g(x+t)] + \frac{1}{2}[g(x+t) - g(x-t)] = g(x+t).$$

Likewise, to get a right-moving wave, we want the functions of $(x+t)$ to cancel out, so we can take $h(x) = -g'(x)$.

**Problem 2.21.** (Equipartition of Energy). Let $u \in C^2(\mathbb{R} \times [0,\infty))$ solve the initial-value problem for the wave equation in one-dimension:

$$\begin{cases} u_{tt} - u_{xx} = 0 & (\text{in } \mathbb{R} \times (0,\infty)) \\ u = g, u_t = h & (\text{on } \mathbb{R} \times \{t = 0\}). \end{cases}$$

Suppose $g, h$ have compact support. The *kinetic energy* is $k(t) \coloneqq \frac{1}{2}\int_{-\infty}^{\infty} u_t^2(x,t)\, dx$ and the *potential energy* is $p(t) \coloneqq \frac{1}{2}\int_{-\infty}^{\infty} u_x^2(x,t)\, dx$. Prove
(i) $k(t) + p(t)$ is constant in $t$,
(ii) $k(t) = p(t)$ for all large enough times $t$.

(i) Let us define the kinetic and potential energy, $k(t)$ and $p(t)$ as given above. We wish to show that the total energy

$$e(t) = k(t) + p(t) = \frac{1}{2}\int_{-\infty}^{\infty} u_t^2 + u_x^2\, dx$$

is conserved, where $u$ is given by d'Alembert's formula,

$$u(x,t) = \frac{1}{2}[g(x+t) + g(x-t)] + \frac{1}{2}\int_{x-t}^{x+t} h(y)\,dy.$$

so that

$$u_t(x,t) = \frac{1}{2}[g'(x+t) - g'(x-t) + h(x+t) + h(x-t)].$$
$$u_x(x,t) = \frac{1}{2}[g'(x+t) + g'(x-t) + h(x+t) - h(x-t)].$$

Let $T > 0$, and let $K \subset \mathbb{R}$ be a compact set large enough so that $\operatorname{supp} g, \operatorname{supp} h$ are both contained in $K$. Then it follows that $u_t^2$ and $u_x^2$ are both supported in $K_T \coloneqq K + [-T, T]$ for all $0 \le t \le T$. Since $u_t^2$ and $u_x^2$ are continuous on $\mathbb{R}$ (and thus on $K_T$) for all $0 \le t \le T$, by the extreme value theorem, there exist finite constants $M$ and $N$ so that for all $x \in K$ and $0 \le t \le T$, $|u_t|^2 \le M\chi_{K_T}$ and $|u_x^2| \le N\chi_{K_T}$, where both dominating functions are integrable since $K_T$ is compact. Thus using the Dominated Convergence Theorem, taking the derivative of $e(t)$ gives us,

$$\begin{aligned}
\dot{e}(t) &= \frac{1}{2}\frac{\partial}{\partial t}\int_{-\infty}^{\infty} u_t^2 + u_x^2\,dx = \frac{1}{2}\frac{\partial}{\partial t}\int_{K_T} u_t^2 + u_x^2\,dx \\
&= \frac{1}{2}\int_{K_T} \partial_t(u_t^2 + u_x^2)\,dx \\
&= \int_{K_T} u_t u_{tt} + u_x \cdot \partial_x u_t\,dx \\
&= \int_{K_T} u_t(u_{tt} - u_{xx})\,dx \qquad \text{(by integration by parts)} \\
&= 0,
\end{aligned}$$

where there is no boundary term since $u_x, u_t$ are supported inside $K_T$. So we have shown that $\dot{e}(t) = 0$ for all $0 \le t \le T$. Since $T > 0$ was chosen arbitrarily, it follows that $\dot{e}(t) \equiv 0$. Therefore, the total energy is constant in $t$.

(ii) What we wish is that in the limit $t \to \infty$,

$$\left|\int_{\mathbb{R}} u_t^2 - u_x^2\,dx\right| \to 0.$$

First, we compute

$$u_t^2 - u_x^2 = h(x+t)h(x-t) - h(x+t)g'(x-t) + h(x-t)g'(x+t) - g'(x-t)g'(x+t).$$

Then

$$\begin{aligned}
\left|\int_{\mathbb{R}} u_t^2 - u_x^2\,dx\right| \le{}& \int_{-\infty}^{\infty} |h(y)h(y-2t)|\,dy + \int_{-\infty}^{\infty} |h(y)g'(y-2t)|\,dy \\
&+ \int_{-\infty}^{\infty} |h(y-2t)g'(y)|\,dy + \int_{-\infty}^{\infty} |g'(y-2t)g'(y)|\,dy.
\end{aligned}$$

Let $K \subset \mathbb{R}$ be a compact set large enough that contains the support of both $g$ and $h$. Then we may write,

$$\begin{aligned}
\left|\int_{\mathbb{R}} u_t^2 - u_x^2\,dx\right| \le{}& \int_{K} |h(y)h(y-2t)|\,dy + \int_{K} |h(y)g'(y-2t)|\,dy \\
&+ \int_{K} |h(y-2t)g'(y)|\,dy + \int_{K} |g'(y-2t)g'(y)|\,dy.
\end{aligned}$$

We will only show that the first term on the right vanishes since the work for the remaining terms is identical. Since $h(u) \to 0$ as $|u| \to \infty$ (by compact support of $h$), we know that pointwise $|h(y)h(y-2t)| \to 0$ as $t \to \infty$. Next since $h$ is continuous, there exists some finite $M > 0$ so that

$|h(y)|, |h(y-2t)| \leq M$ for all $y \in K$. Thus, $|h(y)h(y-2t)| \leq M^2 \chi_K \in L^1(\mathbb{R})$ since $K$ is compact. Hence, by the Dominated Convergence Theorem, one has that the first term vanishes as $t \to \infty$. As we stated before, the remaining three terms also vanish as $t \to \infty$. Thus, we conclude that as $t \to \infty$,

$$\left| \int_{\mathbb{R}} u_t^2 - u_x^2 \, dx \right| \to 0.$$

# 4 Luk

## 4.1 Notes

**Problem N.1.** (Luk, pg. 3) The Einstein vacuum equations describe the propagation of gravitational waves in the absence of matter and take the form $\mathrm{Ric}(g) = 0$, where the Lorentzian metric $g$ is the unknown. Show that in a coordinate system, these equations take the form

$$\mathrm{Ric}(g)_{\mu\nu} = -\frac{1}{2}g^{\alpha\beta}\partial^2_{\alpha\beta}g_{\mu\nu} - \frac{1}{2}g^{\alpha\beta}\partial^2_{\mu\nu}g_{\alpha\beta} + \frac{1}{2}g^{\alpha\beta}\partial^2_{\alpha\nu}g_{\beta\mu} + \frac{1}{2}g^{\alpha\beta}\partial^2_{\beta\mu}g_{\alpha\nu} + F_{\mu\nu}(g,\partial g),$$

where $F_{\mu\nu}(g,\partial g)$ is a function of $g$ and its derivatives.

Let $(M,g)$ be a Lorentzian manifold, and let $\mathrm{Ric}(g)$ denote the Ricci curvature tensor on $M$. Let $(x^j)_{j=1}^n$ denote a local coordinate system, and let $(\partial_j)_{j=1}^n$ be the corresponding vector fields. Let $R$ denote the Riemann curvature tensor. Then if $R$ is given in coordinates by $R = R_{ijk}{}^{\ell}\, dx^i \otimes dx^j \otimes dx^k \otimes \partial_\ell$, then

$$R_{ijk}{}^{\ell} = \partial_i\Gamma^\ell_{jk} - \partial_j\Gamma^\ell_{ik} + \Gamma^m_{jk}\Gamma^\ell_{im} - \Gamma^m_{ik}\Gamma^\ell_{jm},$$

where the $\Gamma^\bullet_{\star\star}$ are the *Christoffel symbols*. One can also show that in coordinates,

$$\Gamma^i_{jk} = \frac{1}{2}g^{im}\left(\partial_k g_{mj} + \partial_j g_{mk} - \partial_m g_{jk}\right).$$

Thus, we then compute,

$$R_{ijk}{}^{\ell} = \frac{1}{2}g^{\ell m}\left(\partial^2_{ki}g_{mj} + \partial^2_{ji}g_{mk} - \partial^2_{mi}g_{jk}\right) + \{\cdots\} - \frac{1}{2}g^{\ell m}\left(\partial^2_{kj}g_{mi} + \partial^2_{ij}g_{mk} - \partial^2_{mj}g_{ik}\right) - \{\cdots\} + \{\cdots\}.$$

Then the component functions of the Ricci curvature tensor in coordinates are,

$$\mathrm{Ric}(g)_{ij} = R_{\ell ij}{}^{\ell} = -\frac{1}{2}g^{\ell m}\partial^2_{\ell m}g_{ij} - \frac{1}{2}g^{\ell m}\partial^2_{ij}g_{\ell m} + \frac{1}{2}g^{\ell m}\partial^2_{im}g_{\ell j} + \frac{1}{2}g^{\ell m}\partial^2_{\ell j}g_{im} + F_{ij}(g,\partial g).$$

Thus to agree with the format presented in the problem, let $\ell = \alpha$, $m = \beta$, $i = \mu$, and $j = \nu$.

**Problem N.2.** (Luk, pg. 18) Let $\phi$ be a solution to the Cauchy problem,

$$\begin{cases} \partial_\alpha(a^{\alpha\beta}\partial_\beta\phi) = F \\ (\phi, \partial_t\phi)\restriction_{\{t=0\}} = (\phi_0, \phi_1). \end{cases}$$

Show that

$$\sup_{t\in[0,T]} \|\phi\|_{L^2(\mathbb{R}^n)}(t) \le C\left(\|\phi_0\|_{L^2(\mathbb{R}^n)} + \int_0^T \|\partial_t\phi\|_{L^2(\mathbb{R}^n)}(t)\, dt\right).$$

Let $\phi$ be a sufficiently regular solution to the aforementioned Cauchy problem. Then the fundamental theorem of calculus tells us that

$$\phi(x,t) = \phi_0(x) + \int_0^t \partial_t\phi(t',x)\, dt',$$

for any $t \in [0,T]$. Thus, we compute,

$$\begin{aligned}
|\phi(x,t)|^2 &\le |\phi_0|^2 + \left| \int_0^t \partial_t \phi(t',x)\, dt' \right|^2 \\
\implies \|\phi\|^2_{L^2(\mathbb{R}^n)} &\le 2\|\phi_0\|^2_{L^2(\mathbb{R}^n)} + 2\int_{\mathbb{R}^n} \left| \int_0^t \partial_t \phi\, dt' \right| dx \\
\implies \|\phi\|_{L^2(\mathbb{R}^n)} &\le 2^{1/2}\|\phi_0\|_{L^2(\mathbb{R}^n)} + 2^{1/2}\left( \int_{\mathbb{R}^n} \left| \int_0^t \partial_t \phi\, dt' \right|^2 dx \right)^{1/2} \\
&\le 2^{1/2}\|\phi_0\|_{L^2(\mathbb{R}^n)} + 2^{1/2}\int_0^t \|\partial_t \phi\|_{L^2(\mathbb{R}^n)}\, dt' \\
&\le 2^{1/2}\left( \|\phi_0\|_{L^2(\mathbb{R}^n)} + \int_0^T \|\partial_t \phi\|_{L^2(\mathbb{R}^n)}\, dt \right),
\end{aligned}$$

where the fourth line follows the Minkowski inequality, and the final line follows from the fact that $\|\partial_t\phi\|_{L^2(\mathbb{R}^n)} \ge 0$.

**Problem N.3.** Consider the general functional inequality,

$$\|\eta\|_{H^s(\mathbb{R}^n)} \le C\|\eta\|^{\theta_1}_{H^{s_1}(\mathbb{R}^n)}\|\eta\|^{\theta_2}_{H^{s_2}(\mathbb{R}^n)}$$

for some $C = C(s_1,s_2,s,n) > 0$, where $0 \le s_1 \le s \le s_2$, $\theta_1+\theta_2 = 1$, and $\theta_1 s_1 + \theta_2 s_2 = s$. Prove this functional inequality by taking the Fourier transform and using Hölder's inequality.

Let $\eta \in H^s(\mathbb{R}^n)$. We recall that

$$\|\eta\|^2_{H^s(\mathbb{R}^n)} \coloneqq \int_{\mathbb{R}^n} (1+|\xi|^2)^s |\hat{\eta}|^2\, d\xi$$

Let $0 \le s_1 \le s \le s_2$, and let $\theta_1, \theta_2$ such that $\theta_1+\theta_2 = 1$ and $\theta_1 s_1 + \theta_2 s_2 = s$. Then we can write,

$$\begin{aligned}
\|\eta\|^2_{H^s(\mathbb{R}^n)} &= \int_{\mathbb{R}^n} (1+|\xi|^2)^{\theta_1 s_1} |\hat{\eta}|^{2\theta_1} (1+|\xi|^2)^{\theta_2 s_2} |\hat{\eta}|^{2\theta_2}\, d\xi \\
&\le \left( \int_{\mathbb{R}^n} (1+|\xi|^2)^{s_1} |\hat{\eta}|^2\, d\xi \right)^{\theta_1} \left( \int_{\mathbb{R}^n} (1+|\xi|^2)^{s_2} |\hat{\eta}|^2\, d\xi \right)^{\theta_2} \\
&= \|\eta\|^{2\theta_1}_{H^{s_1}(\mathbb{R}^n)} \|\eta\|^{2\theta_2}_{H^{s_2}(\mathbb{R}^n)}.
\end{aligned}$$

## 4.2 Example Sheets

**Problem E.1.5.** Given smooth and compactly supported initial data to the linear wave equation in any dimension, use the Fourier representation formula to show that the $\dot{H}^k$ norm is uniformly bounded in time for any $k \ge 1$.

Let $(\phi, \partial_t\phi)\restriction_{\{t=0\}} = (\phi_0, \phi_1)$ be smooth and compactly supported initial data to the linear wave equation in $\mathbb{R}^{1+n}$, and let $\phi$ be the unique solution to the Cauchy wave problem. In the notes, we explicitly showed that

$$\hat{\phi}(t,\xi) = \hat{\phi}_0 \cos(2\pi t|\xi|) + \frac{\hat{\phi}_1}{2\pi|\xi|}\sin(2\pi t|\xi|).$$

We also recall that the norms for the spaces $\dot{H}^k$ (i.e., the homogeeous Sobolev spaces $\dot{W}^{k,2}$) for $1 \le k < \infty$ are given by

$$\|u\|_{\dot{H}^k(\mathbb{R}^n)} \coloneqq \left( \int_{\mathbb{R}^n} |\xi|^{2k} |\hat{u}|^2\, d\xi \right)^{1/2}.$$

Thus, we compute,

$$\begin{aligned}\|\phi\|^2_{\dot{H}^k(\mathbb{R}^n)}(t) &= \int_{\mathbb{R}^n} |\xi|^{2k} \left| \hat{\phi}_0 \cos(2\pi t|\xi|) + \frac{\hat{\phi}_1}{2\pi|\xi|} \sin(2\pi t|\xi|) \right|^2 d\xi \\ &\leq 2\int_{\mathbb{R}^n} |\xi|^{2k} |\hat{\phi}_0|^2 \cos^2(2\pi t|\xi|)\, d\xi + \frac{1}{2\pi^2} \int_{\mathbb{R}^n} |\xi|^{2k-2} |\hat{\phi}_1|^2 \sin^2(2\pi t|\xi|)\, d\xi \\ &\leq 2\|\phi_0\|^2_{\dot{H}^k(\mathbb{R}^n)} + \frac{1}{2\pi^2}\|\phi_1\|^2_{\dot{H}^{k-1}(\mathbb{R}^n)},\end{aligned}$$

for all $t \geq 0$. Thus, we conclude that the $\dot{H}^k$ is uniformly bounded in time for any $1 \leq k < \infty$.

**Problem E.1.6.** (Decay for Klein-Gordon equation) Suppose $\phi$ satisfies the Klein-Gordon equation; i.e.,

$$\Box\phi - \phi = 0$$

on $\mathbb{R} \times \mathbb{R}^n$ with initial data given by $(\phi, \partial_t\phi) \restriction_{\{t=0\}} = (\phi_0, \phi_1) \in C_c^\infty \times C_c^\infty$.

(a) First, show using the Fourier transform that if the solution is sufficiently regular, it is given by the formula

$$\hat{\phi}(t,\xi) = \hat{\phi}_0(\xi)\cos(\sqrt{4\pi^2|\xi|^2+1}\,t) + \frac{\hat{\phi}_1(\xi)}{\sqrt{4\pi^2|\xi|^2+1}} \sin(\sqrt{4\pi^2|\xi|^2+1}\,t).$$

(b) For the $n = 1$ case, prove that $(\sup_x |\phi|)(t) \leq \frac{C}{(1+t)^{\frac{1}{2}}}$, where $C = C(\phi_0, \phi_1) > 0$ is independent of $t$. Note that it suffices (why?) to show that

$$\left| \int_{\mathbb{R}} \exp\left( it\left( \frac{2\pi x\cdot\xi}{t} + \sqrt{4\pi^2\xi^2+1} \right) \right) \hat{\phi}_0(\xi);d\xi \right| \leq \frac{C}{t^{1/2}}$$

for $t$ large. Let

$$\varphi \coloneqq \frac{2\pi x\xi}{t} + \sqrt{4\pi^2|\xi|^2+1}.$$

Show that

$$\frac{\partial}{\partial\xi}\varphi = 0 \implies \xi^2 = \frac{x^2}{4\pi^2(t^2-x^2)}.$$

Show also that

$$\frac{\partial^2}{\partial\xi^2}\varphi \geq \frac{4\pi^2}{(4\pi^2\xi^2+1)^{\frac{3}{2}}}.$$

Now split the integral into $\int_{|\xi^2 - \frac{x^2}{4\pi^2(t^2-x^2)}| \leq \delta} + \int_{|\xi^2 - \frac{x^2}{4\pi^2(t^2-x^2)}| \geq \delta}$, estimate each part with the above observations and optimized in $\delta$ to obtain the desired decay result.

(c) Finally, can you prove that

$$|\phi| \leq \frac{C}{(1+t)^{\frac{n}{2}}}$$

holds for some $C = C(\phi_0, \phi_1) > 0$ for arbitrary dimensions?

(a) Consider the Klein-Gordon equation $\Box\phi - \phi = 0$ on $\mathbb{R}\times\mathbb{R}^n$, with compactly supported smooth initial data. Assume that a solution exists and that it is sufficiently regular. Then taking the Fourier transform of the equation, one obtains

$$-\partial_t^2\hat{\phi} - (4\pi^2|\xi|^2+1)\hat{\phi} = 0 \implies \partial_t^2\hat{\phi} = (4\pi^2|\xi|^2+1)\hat{\phi}.$$

Thus solving the ODE and using the initial data, we obtain (as desired)

$$\hat{\phi}(t,\xi) = \hat{\phi}_0\cos(\sqrt{4\pi^2|\xi|^2+1}\,t) + \frac{\hat{\phi}_1}{\sqrt{4\pi^2|\xi|^2+1}} \sin(\sqrt{4\pi^2|\xi|^2+1}\,t).$$

(b) Rewriting the formula for $\hat{\phi}$ (from (a)) in a more convenient way, we obtain

$$\hat{\phi}(t,\xi) = \exp\left(i\sqrt{4\pi^2\xi^2+1}t\right)\left(\frac{\hat{\phi}_0}{2} + \frac{\hat{\phi}_1}{2i\sqrt{4\pi^2\xi^2+1}}\right) + \exp\left(-i\sqrt{4\pi^2\xi^2+1}t\right)\left(\frac{\hat{\phi}_0}{2} - \frac{\hat{\phi}_1}{2i\sqrt{4\pi^2\xi^2+1}}\right).$$

Thus taking the inverse Fourier transform,

$$\phi(t,x) = I_1 + I_2 + I_3 + I_4,$$

where

$$I_1 = \int_{\mathbb{R}} \exp\left(it\left(\frac{2\pi x\xi}{t} + \sqrt{4\pi^2\xi^2+1}\right)\right)\frac{\hat{\phi}_0}{2}\,d\xi \qquad I_2 = \int_{\mathbb{R}} \exp\left(it\left(\frac{2\pi x\xi}{t} + \sqrt{4\pi^2\xi^2+1}\right)\right)\frac{\hat{\phi}_1}{2i\sqrt{4\pi^2\xi^2+1}}\,d\xi$$
$$I_3 = \int_{\mathbb{R}} \exp\left(it\left(\frac{2\pi x\xi}{t} - \sqrt{4\pi^2\xi^2+1}\right)\right)\frac{\hat{\phi}_0}{2}\,d\xi \qquad I_4 = \int_{\mathbb{R}} \exp\left(it\left(\frac{2\pi x\xi}{t} - \sqrt{4\pi^2\xi^2+1}\right)\right)\frac{\hat{\phi}_1}{2i\sqrt{4\pi^2\xi^2+1}}\,d\xi.$$

As suggested in the problem, it suffices to consider just $t \geq 1$. Since $\phi_0, \phi_1$ are compactly supported smooth functions, their Fourier trannsfoms are Schwartz (thus integrable). This means there exist $M_j > 0$, for $j = 1,2,3,4$ such that $|I_j| \leq M_j$. Thus, taking $M > M_1 + \cdots + M_4$, we have that for all $x \in \mathbb{R}$, $|\phi| \leq M$. Taking $C > 0$ large enough so that $C \geq M \cdot 2^{1/2}$, we have $|\phi(t,x)| \leq C/(1+t)^{1/2}$ for all $t \geq 0$ and $x \in \mathbb{R}$ so that the result follows. Thus we focus on the case for $t \geq 1$. As in the lecture notes, we also only focus on showing the bound for $I_1$. The bound for $I_3$ will follow nearly identically. As hinted in the problem, define

$$\varphi = \frac{2\pi x\xi}{t} + \sqrt{4\pi^2\xi^2+1},$$

and assume that $\partial\varphi/\partial\xi = 0$. I.e.,

$$0 = \frac{\partial\varphi}{\partial\xi} = \frac{2\pi x}{t} + \frac{1}{2}(4\pi^2\xi^2+1)^{-1/2}\cdot 8\pi^2\xi \implies \frac{\pi^2x^2}{t^2} = \frac{4\pi^4\xi^2}{4\pi^2\xi^2+1} \implies \xi^2 = \frac{x^2}{4\pi^2(t^2-x^2)}.$$

Moreover, we compute,

$$\partial_\xi^2\varphi = \frac{4\pi^2}{\sqrt{4\pi^2\xi^2+1}} - \frac{4\pi^2\xi}{(4\pi^2\xi^2+1)^{3/2}}\cdot 8\pi^2\xi = \frac{4\pi^2}{\sqrt{4\pi^2\xi^2+1}} - \frac{16\pi^4\xi^2}{(4\pi^2\xi^2+1)^{3/2}}$$
$$= \frac{4\pi^2}{(4\pi^2\xi^2+1)^{3/2}}.$$

Now we consider the integral

$$I_1 = \int_{\mathbb{R}} \exp\left(it\left(\frac{2\pi x\cdot\xi}{t} + \sqrt{4\pi^2\xi^2+1}\right)\right)\hat{\phi}_0(\xi)\,d\xi = \int_{\mathbb{R}} e^{i\varphi t}\hat{\phi}_0(\xi)\,d\xi.$$

Taylor expanding $\varphi$ around $\xi_0$ (where $\xi_0^2 = x^2/(4\pi(t^2-x^2))$), we have

$$\varphi = \left(\frac{2\pi x\xi_0}{t} + \sqrt{4\pi^2\xi_0^2+1}\right) + \frac{2\pi^2}{(4\pi^2\xi_0^2+1)^{3/2}}(\xi-\xi_0)^2 + \cdots.$$

Therefore,

$$I_1 \approx \hat{\phi}_0(\xi_0)\exp\left(it\cdot\left[\frac{2\pi x\xi_0}{t} + \sqrt{4\pi^2\xi_0^2+1}\right]\right)\cdot\int_{\mathbb{R}}\exp\left(it\cdot\frac{2\pi^2}{(4\pi^2\xi_0^2+1)^{3/2}}(\xi-\xi_0)^2\right)d\xi$$

The latter integral, which we denote by $J$, evaluates to

$$J = \int_{\mathbb{R}}\exp\left(i\cdot\frac{2\pi^2(t^2-x^2)^{3/2}}{t^2}(\xi-\xi_0)^2\right)d\xi = \left(\frac{2\pi}{|c|}\right)^{1/2}e^{i(\pi/4)},$$

where

$$c = \frac{4\pi^2(t^2-x^2)^{3/2}}{t^2}$$

Therefore, one has that

$$|I_1| \approx |\hat{\phi}_0(\xi_0)| \cdot \left(\frac{t^2}{2\pi(t^2-x^2)^{3/2}}\right)^{1/2} \sim \frac{1}{t^{1/2}}$$

where we use the fact that $\hat{\phi}_0$ is Schwartz, and thus, bounded. For $t > 1$, we have $1 + t \le 2t \implies (1+t)^{1/2} \le \sqrt{2}\sqrt{t} \implies \frac{\sqrt{2}}{(1+t)^{1/2}} \ge \frac{1}{t^{1/2}}$. Thus, $|I_1| \lesssim \frac{1}{(1+t)^{1/2}}$, where the constant was dependent only on the initial data. Note that we used the idea of *stationary phase approximations*[1].

(c) We can simply generalize the entire above argument to show that the desired decay rate is satisfied.

**Problem E.1.9.** Show that the bound $\sup_{t,x} |\partial\phi| \le C$ for the solutions to the wave equation in $\mathbb{R}\times\mathbb{R}$ with smooth and compactly supported data is sharp in the following sense: suppose the initial data are given by $(\phi, \partial_t\phi) \restriction_{\{t=0\}} = (\phi_0, \phi_1) \in C_c^\infty \times C_c^\infty$. Then if there exists a non-increasing function $A(t)$ such that $\lim_{t\to\infty} A(t) = 0$ and

$$(\sup_x |\partial\phi|)(t) \le A(t),$$

then $\phi_0 = \phi_1 = 0$.

Let $\phi$ be a solution to the linear wave equation in $\mathbb{R}\times\mathbb{R}$ with compactly supported smooth data, and such that $\sup_{t,x} |\partial\phi| \le C$. We recall d'Alembert's formula, which states that

$$\phi(t,x) = \frac{1}{2}[\phi_0(x+t) + \phi_0(x-t)] + \frac{1}{2}\int_{x-t}^{x+t} \phi_1(y)\, dy.$$

We then compute,

$$|\partial\phi|^2(t,x) = \frac{1}{2}\left[(\phi_0')^2(x+t) + (\phi_0')^2(x-t) + \phi_1^2(x+t) + \phi_1^2(x-t)\right] + \phi_0'(x+t)\phi_1(x+t) - \phi_0'(x-t)\phi_1(x-t).$$

Let $[a,b]^{\text{cpt}} \subseteq \mathbb{R}$ be a compact interval large enough to contain the support of the initial data. Fix an arbitrary point $x_0 \in [a,b]$, and let $x_1(t) = x_0 + t$ and $x_2(t) = x_0 - t$. Then it is easily shown that

$$\lim_{t\to\infty} |\partial\phi|^2(t, x_1(t)) = \frac{1}{2}[(\phi_0')^2(x_0) + (\phi_1)^2(x_0) - \phi_0'(x_0)\phi_1(x_0)]$$

and

$$\lim_{t\to\infty} |\partial\phi|^2(t, x_2(t)) = \frac{1}{2}[(\phi_0')^2(x_0) + (\phi_1)^2(x_0) + \phi_0'(x_0)\phi_1(x_0)].$$

By our hypothesis on the decay rate, we must also have $\lim_{t\to\infty} |\partial\phi|^2(t,x) = 0$ for every fixed $x \in \mathbb{R}$. Thus both of the above limits are zero. This yields $\phi_0'(x_0), \phi_1(x_0) = 0$. But since $x_0 \in [a,b]$ was arbitrary, $\phi_0', \phi_1 \equiv 0$. Thus $\phi_0(x) \equiv c$ is a constant. But since the energy decays to zero as $t \to \infty$, but energy is also conserved, we must have $c \equiv 0$. Thus the conclusion follows.

**Problem E.1.12.** Show using the fundamental solution that solutions to the linear wave equation in $(1+1)$-dimensions; i.e., $\mathbb{R}\times\mathbb{R}$, satisfy

$$\sup_{t,x} |\partial\phi| \le C \sup_x |\partial\phi|(0,x)$$

for some $C > 0$ independent of $\phi$. Show that the corresponding estimate fails in $\mathbb{R}\times\mathbb{R}^3$ (in fact, all dimensions $n \ge 2$!); i.e.,

$$\sup_{\{(\phi_0,\phi_1):\sup_x |\partial\phi|(0,x)=1\}} \left(\sup_{t,x} |\partial\phi|\right) = \infty.$$

This shows that unlike the $L^2$ norm for the derivatives of $\phi$, which is propagated by the equation in all dimensions, the $L^\infty$ norm for the derivative of $\phi$ is not propagated.

[1] https://en.wikipedia.org/wiki/Stationary_phase_approximation

# 5 Random Computations

## 5.1 09/23/2026

**Problem 1.** Analyze the modified energy, $E_p(t)$, defined by: for each $p \in (0,\infty)$, define

$$E_p(t) \coloneqq \|\partial_t\phi\|^p_{L^p(\mathbb{R}^n)}(t) + \|\nabla\phi\|^p_{L^p(\mathbb{R}^n)}(t).$$

For $p=\infty$, define

$$E_\infty(t) = \|\partial_t\phi\|_{L^\infty(\mathbb{R}^n)}(t) + \|\nabla\phi\|_{L^\infty(\mathbb{R}^n)}(t).$$

Let $\phi$ be a sufficiently regular solution to the linear, homogeneous wave equation $\Box\phi = 0$ on $\mathbb{R}^{1+n}$, and for $0 < p < \infty$, let

$$E_p(t) = \int_{\mathbb{R}^n} |\partial_t\phi|^p + |\nabla\phi|^p \, dx,$$

and for $p=\infty$, let

$$E_\infty(t) = \|\partial_t\phi\|_{L^\infty(\mathbb{R}^n)} + \|\nabla\phi\|_{L^\infty(\mathbb{R})}.$$

We have already seen that when $p=2$, $E_p(t)$ is constant in time for all $n \geq 1$. Our goal is to see if this conservation law extends to other $p$. First, let us consider $p=\infty$. We recall the pointwise dispersive estimates,

$$\sup_{x\in\mathbb{R}^n} |\phi(x,t)| \lesssim \frac{1}{(1+t)^{\frac{n-1}{2}}}.$$

Since derivatives commute, we have that $\partial_t\phi$ and $\nabla\phi$ are both solutions to the linear, homogeneous wave equation. So in fact, one has

$$\sup_{x\in\mathbb{R}^n} |\partial_t\phi(x,t)|, \sup_{x\in\mathbb{R}^n} |\nabla\phi(x,t)| \lesssim \frac{1}{(1+t)^{\frac{n-1}{2}}}$$

Therefore for $n \geq 2$,

$$E_\infty(t) = \|\partial_t\phi\|_{L^\infty(\mathbb{R}^n)} + \|\nabla\phi\|_{L^\infty(\mathbb{R}^n)} \lesssim \frac{1}{(1+t)^{\frac{n-1}{2}}} \to 0 \text{ as } t\to\infty,$$

so that $E_\infty(t)$ is not constant in time. Now consider $n=1$. We can easily disprove the conservation law via an example. Consider the Cauchy wave problem,

$$\begin{cases} \Box\,\phi = 0, \\ (\phi, \partial_t\phi) \restriction_{\{t=0\}} = ((-1+x^2)\chi_{[-1,1]}, 0) \end{cases}$$

Then $E_\infty(0) = \|\nabla\phi\|_{L^\infty}(0) = 2$. However, in the period $0 < t < 1/2$, $\|\nabla\phi\|_{L^\infty(\mathbb{R})} > 1$ so that $E_\infty(t) > 2$ for all $t \in (0,1/2)$. Thus $E_\infty(t)$ cannot be conserved. This rules out any energy conservation law for $p=\infty$.

Now let us consider $p=1$,

$$E_1(t) = \int_{\mathbb{R}^n} |\partial_t\phi| + |\nabla\phi| \, dx =: \|\partial_t\phi\|_{L^1(\mathbb{R})} + \|\nabla\phi\|_{L^1(\mathbb{R})}.$$

We will consider the case $n=1$ first. Consider the Cauchy wave problem,

$$\begin{cases} \Box\,\phi = 0, \\ (\phi, \partial_t\phi) \restriction_{\{t=0\}} = ((-1+|x|)\chi_{[-1,1]}, 0) \end{cases}$$

Then $E_1(0) = \|\nabla\phi\|_{L^1(\mathbb{R})} = 2$. Now let $t \in (0,1/2)$. Then

$$\begin{aligned} \|\nabla\phi\|_{L^1(\mathbb{R})}(t) &= \int_{-1-t}^{-1+t} \frac{1}{2}\, dt + \int_{-1+t}^{-t} dt + \int_t^{1-t} dt + \int_{1-t}^{1+t} \frac{1}{2}\, dt \\ &= 2 - 2t. \end{aligned}$$

Likewise, we compute $\|\partial_t\phi\|_{L^1(\mathbb{R})}(t) = 4t$. Therefore, $E_1(t) = 4t+2-2t = 2+2t > 2$. Evidently, $E_1(t)$ is not, in general, conserved in $n = 1$. Now we consider $n \geq 2$. We modify the Cauchy problem from above as follows,

$$\begin{cases} \Box\,\phi = 0, \\ (\phi, \partial_t\phi)\restriction_{\{t=0\}} = ((-1+|x|)\chi_{\{|x|\leq 1\}}, 0), \end{cases}$$

where $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$. Then we compute $E_1(0) = \|\nabla\phi\|_{L^1(\mathbb{R}^n)}(0) = \omega_n$, where $\omega_n$ is the volume of the unit $n$-ball. On the other hand, for $t \in (0, 1/2)$, switching to radial coordinates and using the dispersion of solutions to the wave equation, we can show that the energy exceeds the initial energy. Thus $E_1(t)$ is not conserved for any $n \geq 2$. This rules out any energy conservation law for $p = 1$. Going one step further, we have the following.

**Definition 5.1** (Energy)**.** *Let $n \geq 1$ be a positive integer, and let $\phi$ be a solution to the linear, homogeneous wave equation*

$$\begin{cases} \Box\,\phi = 0, \\ (\phi, \partial_t\phi)\restriction_{\{t=0\}} = ((-1+|x|)\chi_{\{|x|\leq 1\}}, 0) \end{cases} \tag{1}$$

*in $\mathbb{R}^{1+n}$. For each $p \in (0, \infty)$, define*

$$E_p(t) \coloneqq \|\partial_t\phi\|^p_{L^p(\mathbb{R}^n)}(t) + \|\nabla\phi\|^p_{L^p(\mathbb{R}^n)}(t).$$

*For $p = \infty$, define*

$$E_\infty(t) = \|\partial_t\phi\|_{L^\infty(\mathbb{R}^n)}(t) + \|\nabla\phi\|_{L^\infty(\mathbb{R}^n)}(t).$$

**Conjecture.** *Let $n \geq 1$ be a positive integer, and let $\phi$ be a (nontrivial) solution to (1). For all $p \in (0, \infty] \setminus \{2\}$,*

$$\sup_{t\in(0,1/2)} E_p(t) > E_p(0).$$

*If $p = 2$, then $E_p(t)$ is constant for all $t \geq 0$.*

## 5.2 10/07/2026

**Problem 2.** Let $\phi$ be a solution to the linear wave equation $\Box\phi = 0$ in $\mathbb{R}^{1+1}$, with sufficiently regular initial data $(\phi, \partial_t\phi) \restriction_{\{t=0\}} = (\phi_0, \phi_1)$. Then for all $t > 0$ and $p \in [1,2) \cup (2,\infty)$

$$\int_{\Sigma_t} |\partial\phi|^p \, dx \lesssim \int_{\Sigma_0} |\partial x|^p \, dx,$$

where $|\partial\phi|^p \coloneqq |\partial_t\phi|^p + |\partial_x\phi|^p$. Likewise, for $p = \infty$, $\|\partial\phi\|_{L^\infty(\mathbb{R})}(t) \lesssim \|\partial\phi\|_{L^\infty(\mathbb{R})}(0)$.

Let $\phi$ be a solution to the linear wave equation $\Box\phi = 0$ in $\mathbb{R}^{1+1}$ with sufficiently regular initial data $(\phi, \partial_t\phi) \restriction_{\{t=0\}} = (\phi_0, \phi_1)$. First fix $p \in [1,2) \cup (2,\infty)$. d'Alembert's formula gives,

$$\begin{aligned}\partial_t\phi &= F(x+t) + G(x-t),\\ \partial_x\phi &= F(x+t) - G(x-t),\end{aligned}$$

where $F(x) \coloneqq \frac{1}{2}(\phi_0'(x) + \phi_1(x))$, and $G(x) \coloneqq \frac{1}{2}(-\phi_0'(x) + \phi_1(x))$. Then we compute,

$$\begin{aligned}|F(x+t)|^p &\le 2^{-1}|\phi_0'(x+t)|^p + 2^{-1}|\phi_1(x+t)|^p.\\ |G(x-t)|^p &\le 2^{-1}|\phi_0'(x-t)|^p + 2^{-1}|\phi_1(x-t)|^p.\\ |\partial_t\phi|^p &\le 2^{p-1}|F(x+t)|^p + 2^{p-1}|G(x-t)|^p\\ &\le 2^{p-2}\left(|\phi_0'(x+t)|^p + |\phi_1(x+t)|^p + |\phi_0'(x-t)|^p + |\phi_1(x-t)|^p\right)\\ |\partial_x\phi|^p &\le 2^{p-2}\left(|\phi_0'(x+t)|^p + |\phi_1(x+t)|^p + |\phi_0'(x-t)|^p + |\phi_1(x-t)|^p\right)\end{aligned}$$

Thus using translation invariance of the Lebesgue integral on $\mathbb{R}$,

$$\int_{\Sigma_t} |\partial\phi|^p \, dx \le 2^p \int_{\mathbb{R}} |\phi_0'(y)|^p + |\phi_1(y)|^p \, dy = 2^p \int_{\Sigma_0} |\partial\phi|^p \, dx.$$

Now we consider the case $p = \infty$. For any fixed $x \in \mathbb{R}$, one has

$$\begin{aligned}|\partial\phi|(x,t) &= |\partial_t\phi|(x,t) + |\partial_x\phi|(x,t)\\ &\le 2|F(x+t)| + 2|G(x-t)|\\ &\le |\phi_0'(x+t)| + |\phi_1(x+t)| + |\phi_0'(x-t)| + |\phi_1(x-t)|.\end{aligned}$$

Since $t$ is fixed, taking the supremum over all $x \in \mathbb{R}$ and using translation invariance of the $L^\infty$ norm over $\mathbb{R}$, one gets the upper bound,

$$\|\partial\phi\|_{L^\infty(\mathbb{R})}(t) \le 2(\|\phi_0'\|_{L^\infty(\mathbb{R})}(t) + \|\phi_1\|_{L^\infty(\mathbb{R})}(t)) \le 4\|\partial\phi\|_{L^\infty(\mathbb{R})}(0).$$

Thus the proof concludes.

**Problem 3.** Consider solutions to $\Box\phi = 0$ in $\mathbb{R}^{n+1}$. Show that that for all $M > 0$ and $T > 0$, there exist initial data $(\partial_t\phi, \partial_t\phi) \restriction_{\{t=0\}} = (\phi_0, \phi_1)$ such that

$$\int_{\Sigma_0} |\partial\phi|^p \, dx = 1 \qquad \text{and} \qquad \int_{\Sigma_T} |\partial\phi|^p \, dx \ge M,$$

for all $p \in [1,2) \cup (2,\infty]$. (Hint: take $n = 3$, and assume spherical symmetry.)